Komendra Sahu

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Raipur, India

Profile

Engineer who builds and deploys production AI platforms for government departments: capturing requirements, integrating APIs and model services, deploying on Linux/GPU servers, and resolving live issues. Experienced in RAG, hybrid retrieval, and LLM/OCR pipelines.

Experience

CMIT Fellow – Dept. of Electronics & IT, Govt. of Chhattisgarh (deployed with CHiPS) *Present*

- Work directly with **Chhattisgarh government departments** to turn their requirements into deployed GenAI solutions, owning delivery from data onboarding to production support.
- Built **CHiPS-RAG**, a bilingual (Hindi/English) document assistant serving **three department instances** (CHiPS, Finance, General Administration), each with isolated corpus, vector store, login scope, and API keys.
- Deployed and operated services on Linux VMs and **NVIDIA H100 GPUs (Yotta cloud)**: Nginx reverse proxy, Node.js/Express gateway, Flask API, and **vLLM** OpenAI-compatible endpoints serving the LLM and VLM-based OCR; diagnose 502s, port/firewall and model-endpoint failures via logs and health checks.
- Support **Sericulture** and **PHED** departments in preparing their **Detailed Project Reports (DPRs) for CG-AIM**, working directly with department officials to capture requirements.
- Integrated external APIs (Sarvam LLM, speech-to-text, text-to-speech) and self-hosted models behind configurable provider settings, with fallbacks for when a service is unavailable.
- Built the digital platform for **"Seva Mera Yogdaan Utsav"** (2 Oct) *[Website]*: **540+ seva drives across all 33 districts**, **174 organizations** onboarded; featured on the Hon'ble Chief Minister's official Instagram account.

Machine Learning Intern – Unified Mentor *Jan 2026 – Apr 2026 — Remote*

- Built a student segmentation and hybrid course recommender (PCA, K-Means, SVD; Silhouette Score 0.633) for the EduPro platform.

Projects

- **CHiPS RTI Assistant – Legal RAG & PIO Advisory System** *Python, Flask, Node.js, Qdrant, PostgreSQL, BGE-M3, vLLM [Github]*
  Built an end-to-end assistant for **Chhattisgarh RTI** workflows with role-based access for citizens, PIOs, and administrators. A query router (**PostgreSQL / Qdrant / Hybrid**) sends officer-directory lookups to structured data and legal questions to dense+sparse retrieval with cited source PDFs and CIC/CGSIC precedent expansion. Added PIO advisory pipelines with verification safeguards, an **evaluation center** (benchmark datasets, experiments, human review), **Prometheus metrics**, and **pytest/Node** test suites.
- **CHiPS-RAG – Multi-Department Document Assistant** *Python, Flask, Node.js, Qdrant, BGE-M3, Qwen VLM, pgvector, Nginx*
  Designed the ingestion-to-answer pipeline: scraping, **GPU VLM OCR** with a text-quality gate, structure-preserving chunking, **hybrid dense + sparse retrieval with RRF**, optional **BGE reranking**, and grounded answers with source references. Includes admin tools for document review, batch testing, and query-trace diagnostics.
- **DaanUtsav Chhattisgarh '26 – Statewide Giving Platform** *Laravel, PHP, MySQL, Tailwind CSS*
  Built a role-based platform (citizen, NGO, district admin, super admin) for donation listings, volunteer drives, NGO approval queues, and audit logs across 33 districts with an Artisan health-check command.
- **Financial Text Complexity Analytics (NLP & ML)** *Python, FastAPI, React, Random Forest [Github]*
  Built the **Corporate Communication Text Complexity Index (CCTI)** over **200K+ SEC filings** and trained a **Random Forest** model to link textual complexity to stock returns, outperforming standard sentiment-based baselines.

Research & Publications

- **Click Fraud Detection on Resource-Constrained 2-Node YARN Clusters** *ICST 2026 (Accepted) [Link]*
  Spark/YARN pipeline for 100M click records (ROC-AUC 0.9058).
- **Nonlinear Effects of Text Complexity in Corporate Disclosures (CCTI)** *ICTIS 2026 (Accepted) [Link]*
- **Shift-Aware Meta-RL for Robust Auto-Scaling in Serverless Clouds** *ICTIS 2026 (Accepted) [Link]*
- **Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval** *IEEE Xplore, 2025 [Link]*

Technical Skills

Languages: Python, SQL, JavaScript (Node.js), PHP
Deployment & Debugging: Linux, Nginx, systemd, vLLM on H100 GPUs, Docker (basic), Git
ML/AI: RAG, hybrid retrieval, reranking, LangChain, LLM fine-tuning, OCR/VLM, PyTorch, Scikit-learn
Data & Testing: PostgreSQL (pgvector), MySQL, Qdrant, Pandas, NumPy

Education

M.Tech in Data Science & AI, IIIT Naya Raipur *2025–2027* **CGPA: 9.16**

B.Tech in Computer Science & Engg., Govt. Engineering College, Raipur *2021–2025*

GATE 2025 (Computer Science) **Score: 585**